We formalize this identification strategy using a two-stage least squares (2SLS) regression model:

$$1^{\text{st}} \text{ Stage: } \overline{Treat_{\iota} \times post_t} = \beta_0 IV\_weekend_i + \varepsilon_{it}$$
$$2^{\text{nd}} \text{ Stage: } y_{it} = \beta_1 + \beta_2 \overline{Treat_{\iota} \times post_t} + \varepsilon_{it}$$

Here, $\overline{Treat_{\iota} \times Post_t}$ captures exposure to the invitation during the post-invitation period, and $IV_weekend_i$is a binary instrument indicating whether the invitation was sent on a weekend. The unit of analysis remains the reviewer–manuscript pair.

The first-stage regression yields an F-statistic of 1275.8, well above conventional thresholds for weak instruments [23], confirming that weekend timing is a strong predictor of reduced exposure. Full first-stage estimates are reported in SI S4.

Table 2 presents the second-stage estimates from the 2SLS model. In the baseline specification without additional covariates (Model 1), the estimated treatment effect is 0.0288 ($p = 0.005$), meaning that reviewer invitees exposed to the invitation content cited the corresponding author 0.0288 more times post-invitation than their unexposed counterparts. After including the full set of covariates in Model (2) — capturing prior author–reviewer connections, reviewer invitees' characteristics, and journal engagement — the estimated treatment effect remains statistically significant and decreases slightly to 0.0213 ($p = 0.027$). These estimates are robust across specifications and validate our main conclusion: incidental exposure to new research through peer review invitation emails can causally influence subsequent scholarly behavior.

Notably, the 2SLS estimates exceed the baseline difference-in-differences models, suggesting that unobserved confounding likely attenuated earlier results. By addressing endogeneity in exposure, the IV approach offers a more credible estimate of the true causal effect. Together, these findings reinforce the conclusion that peer review is a pervasive conduit through which new and related scientific ideas are adopted not necessarily by the reviewer invitees who come to know the work deeply through formally reviewing a paper, but by reviewer invitees who in the process of declining a review are likely exposed to a concise summary of new ideas.

TABLE 2: INSTRUMENTAL VARIABLE 2SLS

	(1)	(2)
$\overline{\boldsymbol{Treat_\iota \times Post_t}}$	**0.0288***** **(0.0104)**	**0.0213**** **(0.0096)**
# Collaboration for Both		-0.0044 (0.0042)
# Citation Reviewer to Author		0.0061*** (0.0005)
# Citation Author to Reviewer		-0.0003 (0.0002)
Reviewer # Publications (log)		-0.0039*** (0.0002)
Reviewer H-index (log)		0.0063*** (0.0003)
Reviewer # Pub on Journal		-0.0001** (0.0001)
Reviewer # Ref to Journal		-0.0000* (0.0000)
Const	0.0042* (0.0024)	0.0050** (0.0021)
Observations	742,086	742,086
No. clusters	450,737	450,737
R-squared	0.0033	0.1641

Table 2. Instrumental variable estimates of the causal effect of exposure via review invitations on downstream citations. This table reports second-stage estimates from a two-stage least squares (2SLS) regression model. The outcome is the same as Table 1, average annual number of citations from the reviewer invitees to the corresponding author. Exposure is instrumented using a binary indicator for whether the invitation was sent on a weekend, exploiting quasi-random variation in reviewer invitees' email engagement. Model (1) presents the baseline 2SLS estimate without additional covariates: reviewer invitees exposed to invitation metadata cite the corresponding author 0.0288 more times per year ($p = 0.005$) than unexposed reviewer invitees. Model (2) includes prior author–reviewer ties, reviewer invitees' characteristics, and journal engagement history; the estimated effect remains statistically significant at 0.0213 ($p = 0.027$). The first-stage F-statistic is 1275.8 for Model (1) and 1220.48 for Model (2), far exceeding conventional thresholds for weak instruments, supporting the strength of the weekend-sent instrument. These results are robust to model specification and exceed the corresponding difference-in-differences estimates, suggesting that our baseline effects are conservatively estimated. Robust standard errors clustered at the reviewer invitee level are reported in parentheses ($*** \ p < 0.01$, $** \ p < 0.05$, $* \ p < 0.1$).

Attention, Breadth, Depth, Diversity, and Prominence of Citation Behavior Change

We extend our analysis to six complementary dimensions of post-invitation scholarly behavior, each capturing distinct facets of how reviewer invitees may engage with the corresponding author's work post-exposure. Direct Attention captures whether reviewer invitees later cite the specific manuscript they were invited to review, conditional on that manuscript being published. Breadth refers to whether reviewer invitees cite any work by the corresponding author beyond the focal manuscript, indicating a broader engagement with the author's research program. Depth (as measured in our previous analysis) evaluates how extensively the reviewer invitees cite the corresponding author's work within their own publications; higher values suggest more intensive engagement. To account for varying reference list lengths, we also construct a Normalized Depth measure, which adjusts for the total number of references per publication, thus capturing the relative salience of the author's work within each citing paper. Diversity quantifies how many distinct papers by the corresponding author are cited, providing a measure of the range of intellectual influence. Finally, Prominence assesses the prominence of the corresponding author's work within the reviewer invitee's reference list, operationalized using the reciprocal of the earliest citation position in papers as a proxy for conceptual centrality. To measure this, we use the WoS database to extract the citation position within each paper's reference list. See SI S5 for detailed variable definitions.

Figure 4 presents comparisons across six distinct dimensions of reviewer invitees' citation behavior, contrasting the Click-Decline and Auto-Decline groups in the period following the invitation using two-sample t-tests. Group means are shown with 95% confidence intervals, with significance levels annotated. We find that, after the invitation, the Click-Decline group exhibits statistically significant increases across all six dimensions, indicating that peer review promotes not only the diffusion of new and related ideas but also enhances the breadth, depth, diversity, and prominence, with which these ideas are engaged and incorporated into a scholar's research. Corresponding pre-invitation comparisons are presented in SI S5 (Figure S3), confirming the robustness and conservative nature of these results. We further confirm these results using a difference-in-differences (DiD) regression model on the same outcome variables. The estimated treatment effects remain robust and consistent across all six dimensions (SI S5 Table S6).

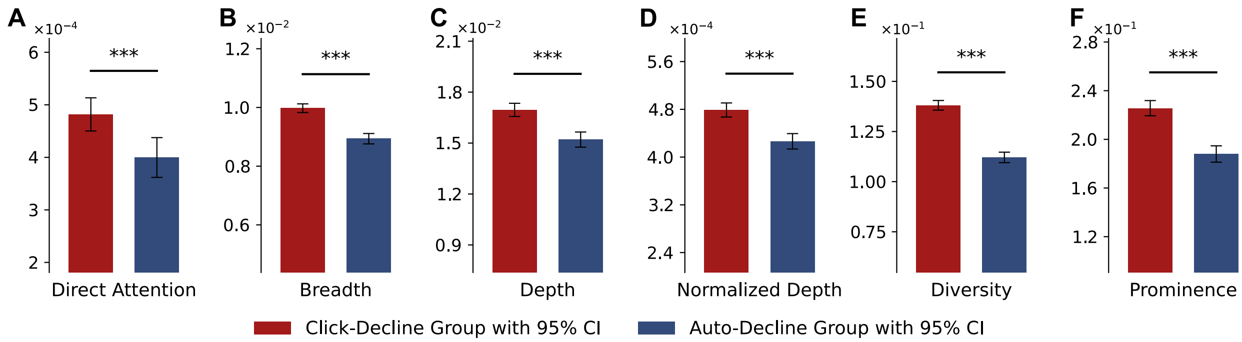

Figure 4. Post-invitation comparisons across six dimensions. We compare the Click-Decline and Auto-Decline groups using two-sample t-tests on six citation-based outcome measures for post-invitation periods. Group means are displayed with 95% confidence intervals, with significance levels and conceptual groupings annotated. Outcome variables include: (A) Direct Attention (proportion of reviewer invitees' publications citing the invited manuscript); (B) Breadth (proportion of reviewer invitees' publications citing any work by the corresponding author); (C) Depth (average number of references to the corresponding author); (D) Normalized Depth (share of references per paper attributed to the author); (E) Diversity (number of unique papers by the author cited); and (F) Prominence (the reciprocal of the earliest reference position of the author's work in reviewer invitees' papers). Color coding indicates conceptual clusters. Following the invitation, Click-Decline invitees subsequently cite the corresponding author with significantly greater direct attention, breadth, depth, diversity, and prominence than Auto-Decline invitees. ($*** \ p < 0.01, \ ** \ p < 0.05, \ * \ p < 0.1$).

Discussion

Our findings demonstrate that peer review is not merely a quality control institution but a powerful, subtle driver of scientific idea diffusion—a previously unrecognized role embedded within this foundational institution of science. Using a novel comparison of click-decline and auto-decline groups, we find that incidental exposure to concise manuscript summaries significantly and durably influences the breadth, depth, diversity and prominence of reviewer invitees' subsequent citations. Given the enormous scale of peer review, with millions of invitations annually, this uncovered effect likely constitutes a substantial yet overlooked mechanism shaping the dissemination and integration of scientific knowledge.

Theoretically, our results broaden existing understanding of how researchers acquire new knowledge. Prior work has primarily emphasized interactive, relationship-oriented channels, such as teams, mentorship, seminars, networks, or leadership [11, 19, 24-27]. Our findings introduce an important complementary dimension: concise, asynchronous exposure to novel ideas through peer review invitations. This channel is particularly valuable in a scientific environment increasingly dominated by informatics-mediated search—which presents popular rather than new knowledge first—and traditional journal publishing, which can slow the presentation of new ideas due to the lengthening of review times and publication lags.

Further, this work speaks to the broader policy questions of how learning takes place in science's increasingly information-rich environment. What is the right combination of deep relationship-oriented learning and concise learning? Does the combination potentially change over a career, team demographics, mentors, fast-moving fields (e.g., AI) vs. slow-moving fields (e.g., math), vanguard researchers that push the boundaries of fields, or among scholars pivoting to new research areas [14, 19, 27, 28]. Moreover, our findings raise novel considerations for journal editors and publishers: as editors select reviewers—an increasingly scrutinized process aimed at fostering diversity and inclusivity—they may also implicitly shape where and how scientific knowledge flows. This underscores an important yet underappreciated editorial responsibility, suggesting that strategic reviewer selection could influence not only manuscript quality assessment but also the broader diffusion and integration of new ideas across scientific communities. Lastly, LLMs are a promising addition to the review process and could provide new technological avenues for presenting knowledge in new and useful ways [29].

Several limitations merit consideration. While we have carefully identified the effect of peer review on idea diffusion, questions remain about how concisely summarized information is most effectively communicated. With research increasingly demonstrating that the semantic underpinnings of papers and grants work hand-in-hand with a paper's robust statistics in showing the merits of good ideas to reviewer invitees [30], it was beyond this study's scope to examine how variation in the presentation of concisely summarized new ideas may impact reviewer invitees' reactions [31]. Related to the linguistic nature of science, while our natural experiment was high-scale and based on a standardized science-wide peer-review system that suggests that the findings are extrapolatable to other journals, future research may further confirm and generalize our findings by involving diverse journals and disciplines.

At a time when peer review faces increasing scrutiny for inefficiencies, high costs, and reviewer fatigue, our findings highlight significant hidden benefits. Peer review emerges as a unique and